

JIANNA EVANS

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EDUCATION

Fayetteville State University

Bachelor of Science

Major in Computer Science; Minor in Mathematics

Cumulative GPA: 3.82

Fayetteville, NC

Expected May 2026

PUBLICATIONS

Aryal, Girja Mani, Kandel, Krisha Prasad, Adhikari, Menuka, Evans, Jianna, Dangol, Hisila, Poudel, Madan, Pokharel, Sushil, Shrestha, Ramdeep, Gautam, Bhoj, and Neupane, Bhanu Bhakta. (2024). Understanding the material properties of alkali-treated lignocellulose fiber obtained from high-altitude Lokta bushes. Retrieved from <https://par.nsf.gov/biblio/10526738>. Biomass Conversion and Biorefinery . Web. doi:10.1007/s13399-024-05278-x.

RESEARCH EXPERIENCE

Physics & Chemistry Department – Fayetteville State University

Materials Science Undergraduate Researcher (Lab PI: Dr. Bhoj Gautam)

Fayetteville, NC

Aug 2022 – Present

- Developed and optimized synthesis parameters for the Ta₄C₃ MXene nanomaterial.
- Fabricate MXene-cellulose biocomposites for wearable semiconductor applications.
- Conducted exploratory data analysis on binary oxides, filtering candidates based on bandgap and phonon stability for machine learning applications.
- Prepared quantum dots for Nb₄C₃ and quantum dot-polymer composite systems.

Computer Science & Mathematics Department – Fayetteville State University

Computer Science Undergraduate Researcher (Lab PI: Dr. Albert Chan)

Fayetteville, NC

Jan 2025 – Present

- Studied the architecture and source code of the Java compiler to understand core compilation components and workflows.
- Built and tested custom compiler versions with modified language features and syntax rules.
- Developed a tailored Java compiler enforcing compliance with the AP Java subset by restricting unsupported features.
- Reduced OpenJDK compiler dependencies and functionality to create a lightweight, educational-focused compiler.

Integrative Bioinformatics Support Group (NIEHS)

National Institute of Environmental Health Sciences Scholar

Durham, NC

Jun 2023 – Jan 2024

- Conducted a comprehensive literature review on genomic analysis pipelines, focusing on 3D genome structure and spatial data integration.
- Designed a customizable wrapper workflow for Hi-C data analysis and three-dimensional genomic interpretation.
- Integrated established tools (HOMER, HiC-Pro, Juicer) to assess A/B compartments, TADs, chromatin loops, and significant chromosomal interactions.
- Presented a proposed project plan to NIEHS researchers and bioinformatics professionals, incorporating technical feedback and project refinement.

INDEPENDENT PROJECTS

YOLOv-Based Object Detection and Video Analytics (Ongoing)

2025

- Implemented YOLOv models for object detection, classification, and counting in video data, including traffic analysis and pedestrian movement heatmaps in mall environments. Currently extending the pipeline to support real-time camera integration and live inference.

MedMNIST Dataset Image Classifier (Ongoing)

2025

- Developed image classification pipelines on the MedMNIST dataset using traditional machine learning methods, including SVM, KNN, Logistic Regression, Decision Trees, and Random Forests. Currently comparing these approaches against deep learning models such as DenseNet to benchmark classification results across model types.

PRESENTATIONS

<i>“Effect of Laser Wavelength on Raman Features of Ti_3C_2Tx MXene”</i>	Raleigh, NC
Poster Presentation at the NC Space Symposium at NC State University	2023
<i>“Developing a Customizable Wrapper for Hi-C Data Analysis”</i>	Durham, NC
Poster and Oral Presentation at NSCP Scholar Symposium at the NIEHS	2023
<i>“Impact of Fluorination of Conjugated Polymer on Aggregation and Charge Photogeneration”</i>	Durham, NC
Poster Presentation at Southeastern Regional Meeting (SERMACS) at Duke University	2023
<i>“Characterizing Structural and Thermal Properties of Nb_4C_3Tx MXenes”</i>	Durham, VA
Poster Presentation at Electronic Materials Conference (EMC) at Duke University	2025
<i>“Structural and Compositional Properties Ta_4C_3Tx MXene”</i>	Denver, CO
Poster Presentation at the APS Global Physics Summit	2026

TECHNICAL SKILLS

Languages: Python, C++, SQL, , Java, R Programming, Pearl, Haskell

Frameworks: PyTorch, TensorFlow, OpenCV, VASP, VESTA

Tools: GitHub, Git, Linux, Jupyter Notebook, Adobe Creative Cloud, Excel, Google Collab

Materials Characterization Techniques: XRD, Raman, FTIR, Fluorescence/Emission Spectroscopy, TCSPC Lifetime Measurements, DSC, TGA

EDUCATIONAL & CLINICAL EXPERIENCE

Computer Science Tutor (Fayetteville State University) 2025 - Current

Provided tutoring for CSC 120, an introductory CS course, focusing on algorithm design, data flow, and computational reasoning. Supported classroom instruction by assisting students with coursework and problem-solving during assignments.

Registered Behavior Technician (RBT) 2024 – Current

Administered individualized, evidence-based ABA therapy to children with developmental disabilities in a clinical setting. Implemented data-driven interventions to support behavioral regulation, communication, and functional skill development.